

Exploratory Data Analysis

Amazon ML Challenge 2026 — Business Entity Resolution

Scope: all six source files and the training ground truth (~24M records). Every figure in this report was computed from the provided data by code/eda.py.

Key findings

- **Heavily linked data.** Only 5.6% of Source 1 entities are singletons; the average entity has 3.46 matches (up to 11). Recall matters as well as precision.
- **Every S2/S3 record belongs to at most one S1 entity** (7.64M matched IDs, all unique). About 26% of S2/S3 records match nothing and act as distractors.
- **Country agrees in 100% of true pairs**, so it is a free blocking key. The test set adds France (15% of test S1), which has no labels in training.
- **Name alone is unreliable.** 50% of S1 entities share a normalised name with another S1 entity, and about 6% of non-Indic true matches have a name similarity below 60 (a brand, an acronym or a domain).
- **Distractors are adversarial near-copies:** same name and street, a house number off by exactly 1-5, 7, 9 or 11, and a changed legal suffix. House-number equality (87.7% of true pairs vs 1.1% of hard negatives) and legal-suffix equality (65% vs 1.5%) are the strongest precision signals.
- **Simple exact blocking keys reach only 79.8% pair recall** (an upper-bound macro F0.5 of about 0.91). Fuzzy or embedding-based candidate generation is required.
- **India names in S2/S3 are 13-24% non-Latin script** (Devanagari, Telugu, Tamil, Kannada, Bengali): transliterations of the English name. Transliteration is a necessary preprocessing step.
- **No leakage:** entity IDs and row order are uncorrelated with matches, and train and test share no IDs.

1. Problem and evaluation

Source 1 is a deduplicated reference list of businesses. For every S1 entity we must return all S2 and S3 records that refer to the same real-world business, which may be none, one or many. The three sources share no identifiers; each record has only *business_name*, *business_address* and *country*.

The metric is **F0.5, computed per S1 entity and then macro-averaged** over all entities, singletons included. $F0.5 = 1.25 \cdot P \cdot R / (0.25 \cdot P + R)$ weights precision twice as heavily as recall. A singleton scores 1.0 for an empty prediction and 0.0 for any prediction. Two files are submitted: *matching_results.tsv* (scored) and *candidate_pairs.tsv* (used to audit blocking recall and reduction ratio). Constraints: MIT/Apache-2.0 models of at most 8B parameters, and no external data lookup (including geocoding).

Implication. Because singletons are only 5.6% and the average entity has 3.5 matches, predicting only the most confident match leaves a lot of F0.5 on the table. With perfect precision and 50% recall an entity scores 0.83; with 100% recall it scores 1.0.

2. Dataset overview

File	Rows	US	India	France	Name mean len	Addr mean len
train_source1	2,206,821	1,323,633	883,188	—	24.0	52.1
train_source2	5,034,616	3,016,817	2,017,799	—	25.1	46.2
train_source3	5,285,603	3,170,056	2,115,547	—	25.2	46.7
test_source1	1,732,544	663,106	809,986	259,452	23.8	57.2
test_source2	4,887,273	1,871,330	2,312,565	703,378	25.7	50.4
test_source3	5,082,316	1,945,701	2,405,000	731,615	25.7	48.7

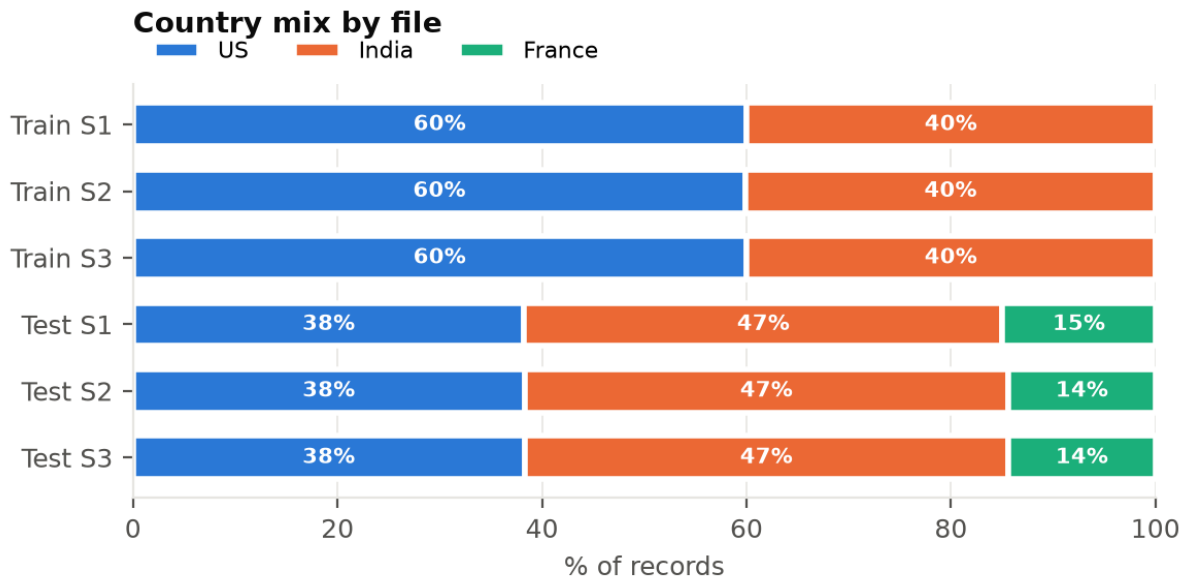


Figure 1. The country mix shifts between train and test: France (unseen in training) appears only in the test files. S2 and S3 are each about 2.3-2.9× the size of S1.

File integrity: read every column as a string with `keep_default_na=False`; otherwise names such as "NULL" or "NA" are silently turned into missing values.

3. Ground-truth structure

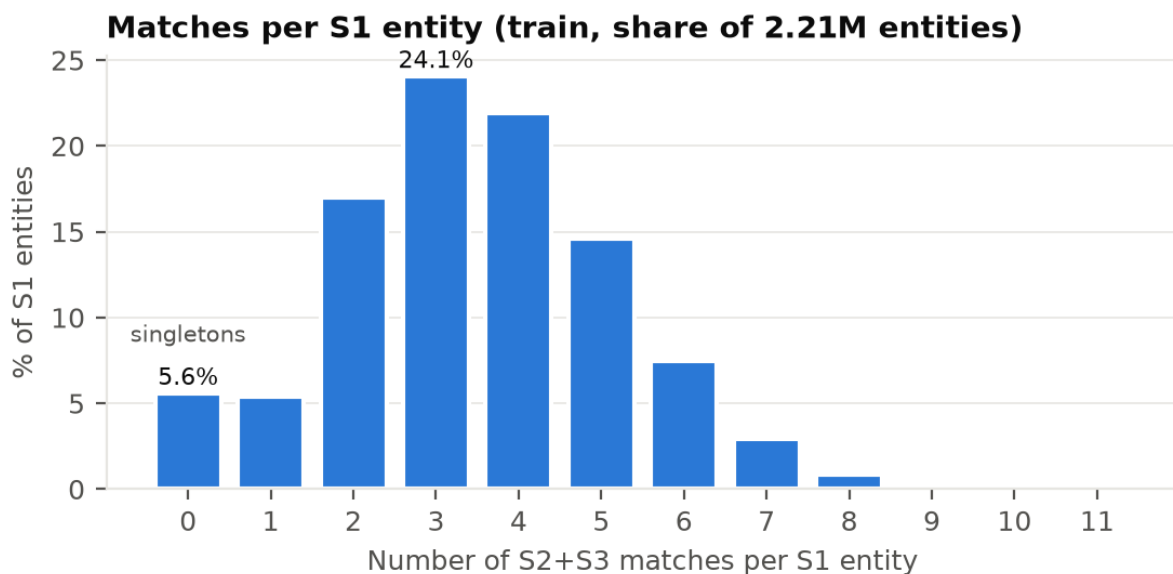


Figure 2. Distribution of matches per S1 entity; only 5.6% are singletons.

Statistic	Value
Singleton S1 entities	123,247 (5.58%); India 5.59%, US 5.58%
Mean matches per S1	3.46 (India 3.46, US 3.46)
Max matches from S2 / S3 for one entity	5 / 6, so S2 and S3 are not deduplicated
Total true pairs	7,638,365
S2/S3 IDs matched to more than one S1	0
Share of S2 / S3 records matched to some S1	73.4% / 74.6%
Country agreement in true pairs	100.0%
Correlation of numeric ID / row position between pair members	0.0001 / 0.0001 (no leakage)

4. Field quality and script mix

File	Empty address	Literal NULL / None in address	Non-ASCII name	Devanagari name	Other Indic name	US address all-caps
train S1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
train S2	3.4%	2.6%	15.2%	5.4%	4.1%	93.5%
train S3	3.3%	2.5%	11.5%	3.0%	2.3%	3.5%
test S1	0.0%	0.0%	2.4%	0.0%	0.0%	0.0%
test S2	2.6%	2.2%	19.0%	6.3%	4.9%	93.4%
test S3	2.7%	2.1%	14.5%	3.6%	2.7%	2.8%

Non-ASCII names in test S1 are French accented names. S1 is otherwise clean, title-cased Latin text.

- **Postal codes are nearly absent:** 6-digit PINs appear in 0.0% of India S1 addresses and 5-digit ZIPs in 11% of US S1 addresses, so postal-code blocking is not viable.
- **India addresses are long and unstructured** (4.7 commas on average vs 2.2 for the US), and 11% of India S1 addresses contain landmark phrases (Near, Opp, Behind, Next to).
- **Empty addresses concentrate in true matches** (4.4% of matched S2/S3 records vs 0.3% of unmatched ones). For these pairs the model must rely on the name.
- **Duplicate names within S1 are common:** 30% of S1 rows repeat an exact raw name, and 50% share a normalised name.

5. Noise patterns (observed examples)

Real S1 → S2/S3 pairs taken from the ground truth (see step 3 output for more):

Pattern	S1 value	Matched S2/S3 value
Case, prefix and legal suffix	CH Dynamic Auto Glass	The CH Dynamic Auto Glass · CH Dynamic Auto Glass Inc.
DBA / trade name	CH Dynamic Auto Glass	Rizatavo Co doing business as CH Dynamic Auto Glass
Injected accents, punctuation, &	Bureau of Parks and Recreation	Bureau óf Parks and-Recreation · Bureau of Parks & Recreation Authority
Typos	Blue Alliance / First Networks Inc	Blue Alliatnae / FIRST NELWKSIB INC
Domain / handle as name	Phoenix Vidyalaya · Youth Unified Program	phoenixvidyalaya.com, ph0enixvidyalaya.com · #youthunified
Junk tokens	Phoenix Vidyalaya	Phoenix Center (ID: 47290) · Dr Phoenix Vídyalaya
Word-order change	Jay It Private Limited	Private Jay It Limited
Script change (India)	Premier Foundation Private Limited	[same name written in Devanagari]
Unrelated name	White Nuclear Inc	Jaxorbi (identical address)

Pattern	S1 value	Matched S2/S3 value
House-number noise	11265 Sunrise Gold Circle	011265 SUNRISE GOLD CIRCLE · ##15140 ELDERFLOWER LANE
Component reorder, full state	5212 Tinkers Creek Place, Clinton, MD	MD, 5212 TINKERS CREEK PLACE, CLINTON · ..., Clinton, Maryland
City alias / misspelling	Pune, Maharashtra · Penfield, NY	Poona, Pune, MH · Pennfield, New York
State in native script	..., Karnataka	..., [Karnataka in Kannada script] / KA
Partial / empty address	Sno 32/2/1 Hno 1048, Gulabnagar, ..., Pune	BLOCK A-825 SNO 32/2/1 HNO 1048, PUNE, ... · (empty)
French abbreviations (test)	Rue de Dieppe, Lille	63 R. DE DIEPPE, LILLE · 51 R DE LA PLANCHE AU GUE

Rate among true pairs (200k sample)	India S2	India S3	US S2	US S3
Raw name exactly equal	2.8%	2.7%	6.0%	5.9%
Normalised name exactly equal	42.7%	44.7%	56.7%	52.7%
Name in Indic script	23.7%	13.2%	0.0%	0.0%
First house number equal	66.3%	65.3%	76.5%	77.4%
Empty address	3.9%	4.1%	4.9%	4.6%
Domain-style name	3.5%	3.8%	4.6%	4.4%
DBA pattern	0.0%	1.6%	0.0%	2.1%

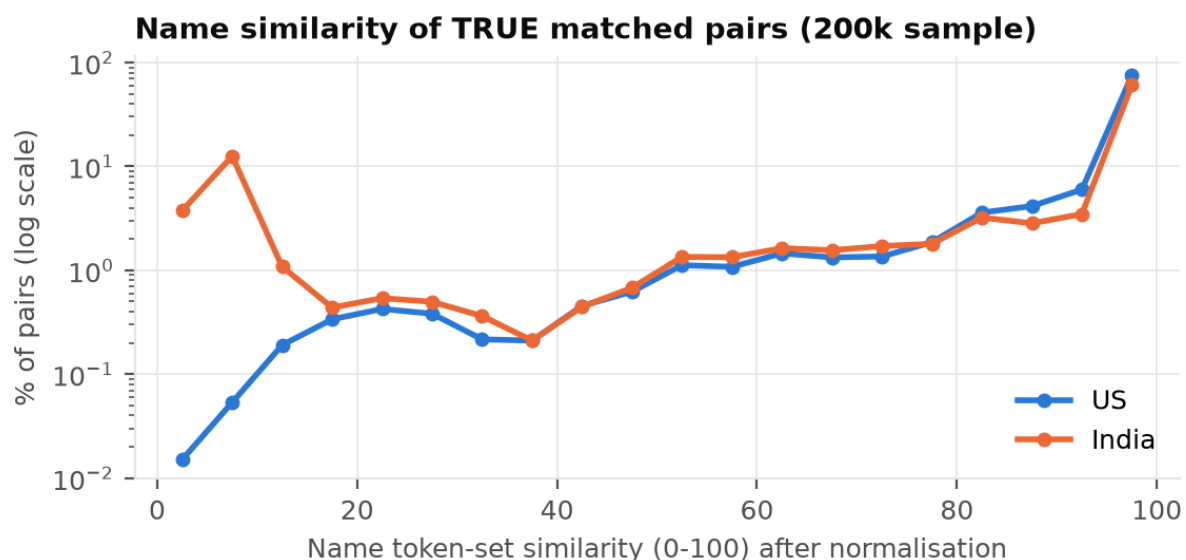


Figure 3. Name similarity of true pairs. India has a heavy low tail, mostly Indic-script names that raw string metrics cannot compare. About 6% of non-Indic true pairs score below 60.

Similarity (true pairs)	India S2	India S3	US S2	US S3
Name token-set, 5th percentile	5	6	60	59
Name token-set, median	100	100	100	100
Address token-set, 5th percentile	60	43	53	54
Address token-set, median	95	91	93	86

6. Hard negatives: how distractors are built

US S2/S3 records were joined to every US S1 entity with the same normalised name, and pairs with an address token-set similarity of at least 70 were kept (176,897 pairs, 82% of them true). The rest are near-copies that differ only in small details:

Distractor (S2)	Closest S1 (not its true entity)
Schulman's Industries Center Corp · 18641 MANN LN, FAIRHOPE, AL	Schulman's Industries Center · 18634 Mann Lane, AL, Fairhope
Golden Electronics Dynamics, Llc · 142 LINCOLN ST, LEOMINSTER, MA	Golden Electronics Dynamics, Corp · 135 Lincoln Street, Leominster, MA
Mann Safe Activate Inc · 615 Eye St, HARLINGEN, TX	Mann Safe Activate LLC · 606 Eye Street, Harlingen, TX
Bobby Banuelos Quality Diagnostics Ltd · 2813-YANCEYVILLE STREET	Bobby Banuelos Quality Diagnostics Inc · 2811 Yanceyville Street, Unit B

House-number difference: the distractor fingerprint (US, same-name & similar-address pairs)

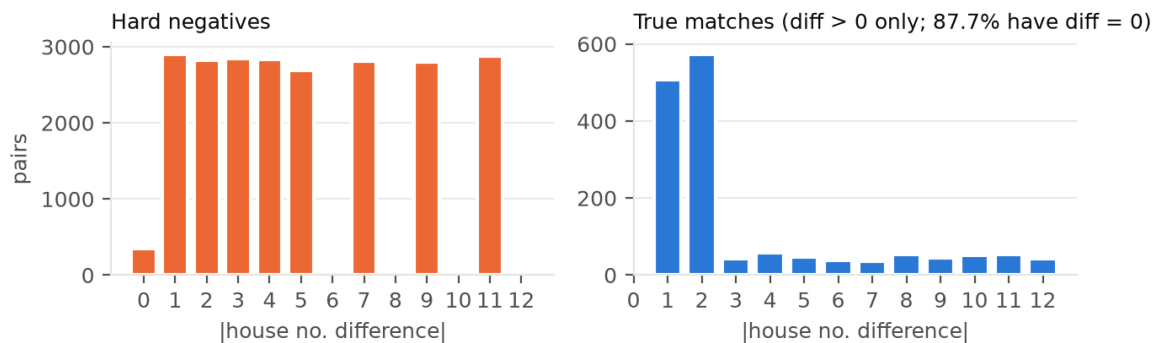


Figure 4. Among hard negatives the house-number difference is almost always exactly 1, 2, 3, 4, 5, 7, 9 or 11; 6, 8, 10 and 12 almost never occur. True pairs mostly keep the same number; their non-zero differences are mostly 1-2 (digit noise) or large (a unit or plot number parsed as the house number).

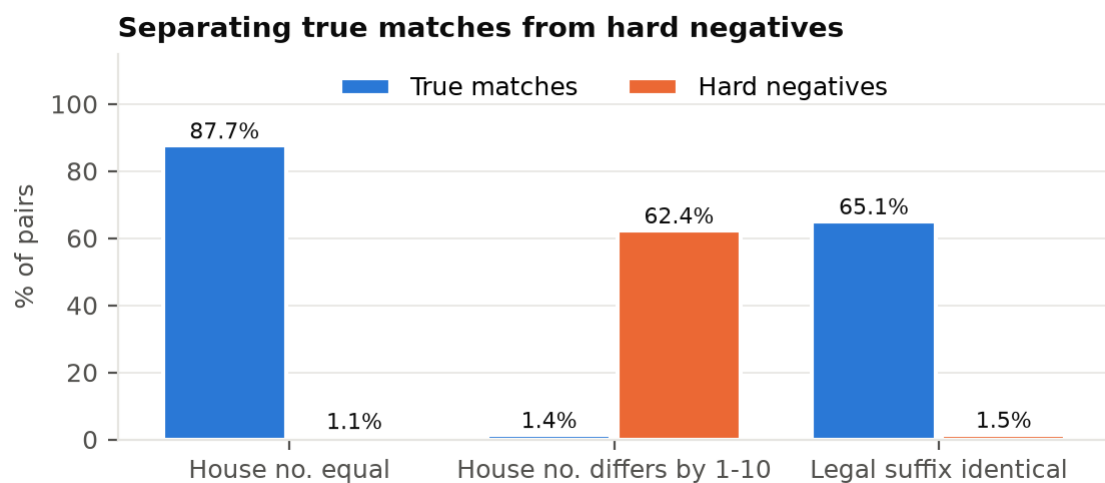


Figure 5. Exact house-number equality and exact legal-suffix equality separate the classes well. Hard negatives almost always change the legal suffix (LLC → Corp, Inc → Ltd).

Implication. A name-plus-address fuzzy score will rank these distractors as matches, and on a singleton a single false merge costs the full 1.0. Feature engineering must include a robust house-number parser, the absolute difference and membership of the difference in {1,2,3,4,5,7,9,11}, and legal-suffix extraction with equality and compatibility flags. A changed suffix alone is not decisive: 35% of true pairs also change it.

7. Blocking (candidate generation) analysis

Within-country exact keys were evaluated against all 7.64M true training pairs. Addresses were lower-cased, accents removed, common street words abbreviated and leading zeros stripped.

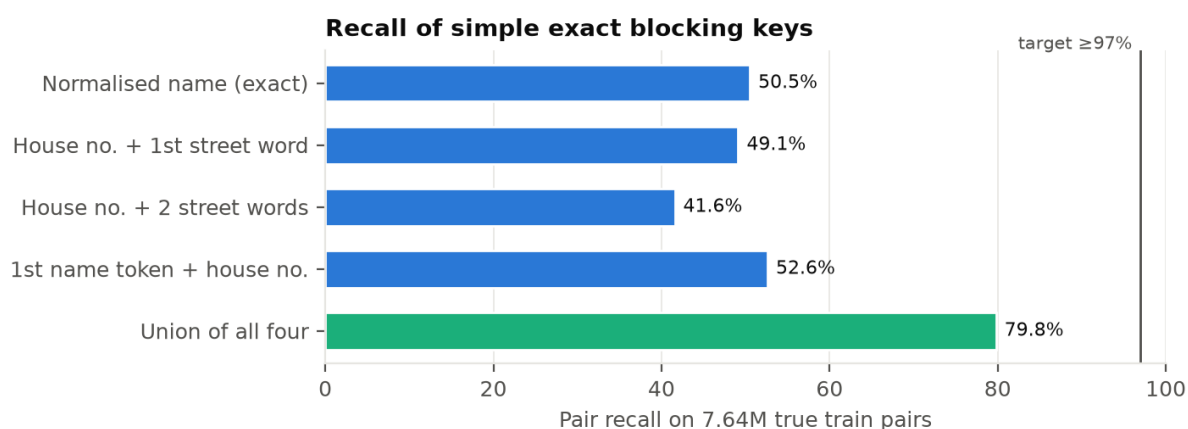


Figure 6. Pair recall of simple exact keys.

Key (within country)	Pair recall	Candidate pairs	Per S1	Largest block
Normalised name	50.5%	106.7M	48.3	970,700
House no. + first street word	49.1%	173.5M	78.6	8,899,035
House no. + two street words	41.6%	13.6M	6.1	543,099
First name token + house no.	52.6%	22.6M	10.2	191,700
Union of all four	79.8%	—	—	—

With the name, house+street and name-token+house keys and a perfect matcher, the best achievable macro F0.5 is **0.909**. Predicting an empty list for every entity scores 0.056.

Implication. The blocking stage caps recall and must combine several methods: (a) the exact keys above; (b) TF-IDF character n-gram (3-4) nearest neighbours on name + address within each country; (c) nearest neighbours on a small multilingual text encoder (MIT/Apache) to catch Indic-script and unrelated-name cases, which also generalises to France. Target at least 97% pair recall at about 20-50 candidates per S1, measured on a held-out split. Frequent-token keys need block-size caps.

8. Recommended preprocessing

Reading and cleaning

- Read with `sep="\t", dtype=str, keep_default_na=False, quoting=csv.QUOTE_NONE`.
- Treat NULL, None, N/A and null inside addresses as empty tokens; add a flag for an empty address.
- Apply Unicode NFKD, strip combining accents and collapse repeated whitespace.

Names

- Lower-case; & → "and"; remove junk prefixes and suffixes (--, ***, #, The, Dr, Smt, "(ID: n)", "| www...", brackets).
- Split DBA names ("X doing business as Y", "dba", "d/b/a", "t/a") and keep both parts.
- Recover words from domains and handles (firstnetworks.com → firstnetworks) and use them for character-level comparison.
- **Extract the legal suffix as a separate field** rather than just deleting it, and map variants to one form: Pvt/Private, Ltd/Limited, L.L.C./LLC, Corp/Corporation, Co/Company, plus SARL/SAS/SASU/EURL/SCI/SA for France.
- Transliterate Indic scripts to Latin, either with an MIT/Apache transliteration library or with a character mapping learned from the aligned training pairs.

Addresses

- Parse into house/plot number, street, unit, city, state and landmark. Strip leading zeros and "##"/"###" from numbers.
- Abbreviation tables per country: US (Street→St, Road→Rd ...), India (H.No, Plot No, Opp, Nr), France (Rue→R, Boulevard→Bd, Avenue→Av, bis/ter).
- Map state names, codes and native-script state names to one canonical code, using a dictionary learned from the data.
- Build a city alias table from training pairs (Poona↔Pune, Calcutta↔Kolkata, "City of X"↔X) and use typo-tolerant city matching.
- Compare addresses as sorted component sets, since component order varies.

Generalising to France

- Treat country as an open set. Every normalisation rule should be country-generic, with an optional country-specific table on top.
- Prefer features that do not depend on country (numeric equality, character n-gram similarity, embedding similarity) so the matcher transfers to unlabelled France.

9. Suggested modelling pipeline

Stage	Approach
Blocking	Union of exact keys, TF-IDF character n-gram kNN and multilingual-embedding kNN (FAISS on GPU), all within country; cap block sizes. Write the final union to candidate_pairs.tsv.
Pair features	Name: token-set / Jaro-Winkler / TF-IDF cosine, legal-suffix equal, domain and DBA flags, name frequency in S1. Address: house number equal, absolute difference, difference in {1,2,3,4,5,7,9,11}, street / city / state / unit similarity, empty flags. Source (S2/S3), country, embedding cosine.
Matcher	Gradient-boosted trees (LightGBM/XGBoost) trained on candidate pairs, with hard negatives taken from the blocking output. Optionally, a small fine-tuned cross-encoder for uncertain pairs.
Post-processing	Assign each S2/S3 record only to its best-scoring S1 (the ground truth is one-to-many). Tune the threshold for macro F0.5 on validation. Add transitive support: an S3 record almost identical to an accepted S2 record is also accepted.
Validation	Hold out about 10% of S1 entities with all their candidates, compute macro F0.5 exactly as specified, and report blocking recall and reduction ratio separately.

Appendix: method notes

- Code: code/eda.py (this report is generated by it). Results are saved to results.json next to the PDF.
- Name normalisation: NFKD accent removal, lower-case, &→and, punctuation removed, "(ID: n)" removed, legal and generic tokens dropped (inc, llc, ltd, pvt, private, corp, co, the, holdings, group, sarl, sas ...).
- Similarity statistics use a 200,000-pair random sample of true pairs (seed 0). The hard-negative analysis uses a 400,000-record US sample of S2 ∪ S3 joined to S1 on normalised name.
- House number is the first integer in the address string, which explains the long tail of large differences for true pairs where a unit or plot number comes first.
- Examples are verbatim records from the training data. Indic-script strings are described in words because the report font lacks those glyphs.